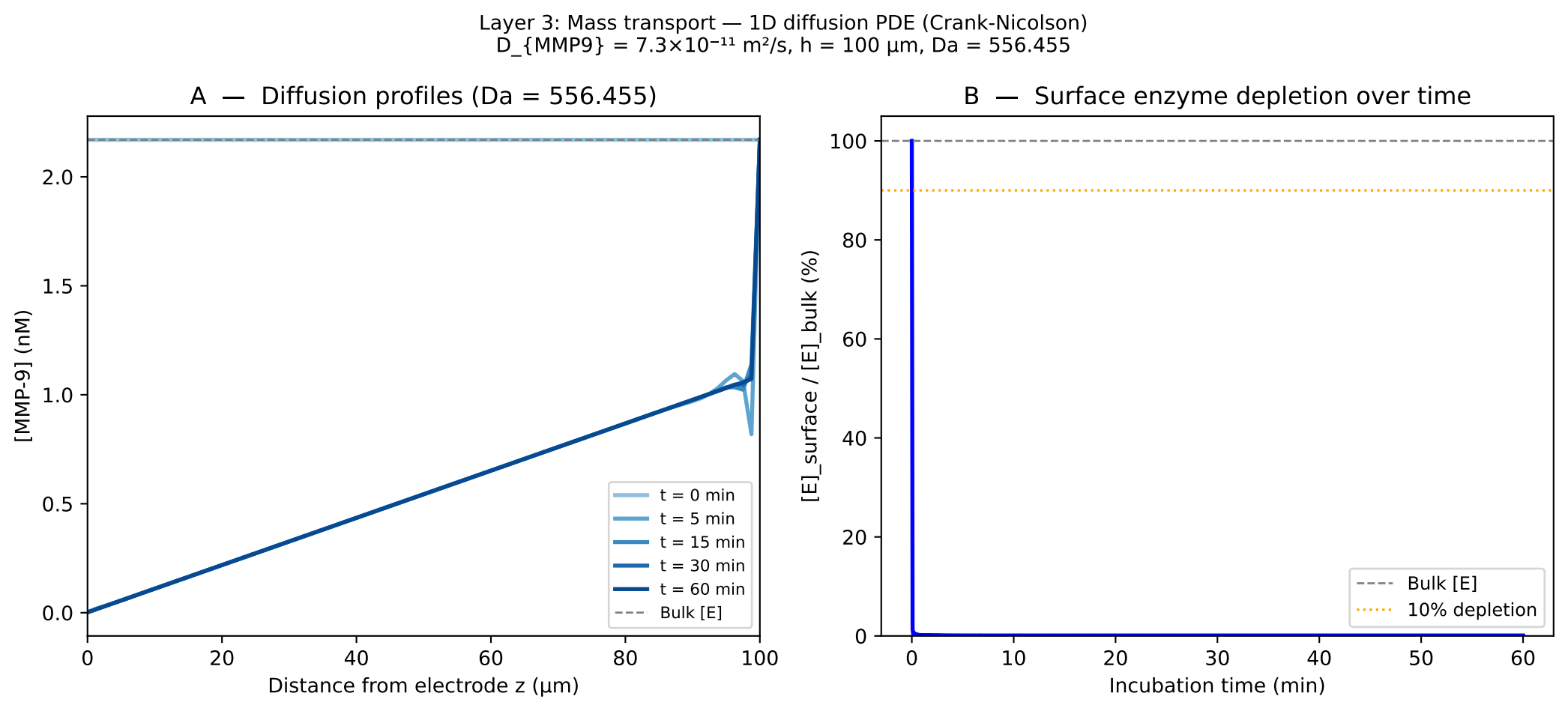
Layer 3: Mass transport — 1D diffusion PDE (Crank-Nicolson)
D_{MMP9} = 7.3×10⁻¹¹ m²/s, h = 100 µm, Da = 556.455
A — Diffusion profiles (Da = 556.455)
[MMP-9] (nM)
Distance from electrode z (µm)
t = 0 min
t = 5 min
t = 15 min
t = 30 min
t = 60 min
Bulk [E]
B — Surface enzyme depletion over time
[E]_surface / [E]_bulk (%)
Incubation time (min)
Bulk [E]
10% depletion